Experiment 2

Aim : Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets

Data Source : (Used Car Price Prediction Dataset)

- Dataset Type: Structured tabular dataset
- File Format: CSV (Comma Separated Values)
- Total Records (Rows): 7,400
- Total Attributes (Columns): 29
- Domain: Used Car Sales / Automobile Marketplace Data
- Dataset Link

Theory :
Regression is a supervised machine learning technique used to predict a continuous output variable based on one or more input features. In real-world datasets (such as housing prices, sales forecasting, medical analysis, or student performance), multiple input variables influence the target value.

To handle such problems effectively, we use:

- Multiple Linear Regression
- Ridge Regression (L2 Regularization)
- Lasso Regression (L1 Regularization)

These models help in prediction, reducing overfitting, and handling multicollinearity.

Multiple Linear Regression

Multiple Linear Regression is an extension of simple linear regression where more than one independent variable is used to predict a dependent variable.

Mathematical Model

Where:

- YY = Dependent variable
- $X1,X2,...Xn$X1,X2,...Xn = Independent variables
- $\beta 0$β0 = Intercept
- $\beta 1,\beta 2...\beta n$β1,β2...βn = Coefficients
- εε = Error term

Working

- The model minimizes the Sum of Squared Errors (SSE).
- Coefficients are calculated using the Ordinary Least Squares (OLS) method.

Limitations

- Prone to overfitting.

- Sensitive to multicollinearity (high correlation among features).

Ridge Regression (L2 Regularization)

Ridge Regression is a regularized version of multiple regression that adds a penalty term to reduce model complexity and prevent overfitting.

Mathematical Formula

$Loss=SSE+\lambda\sum\beta i2$Loss=SSE+λ∑βi2

Where:

- λλ (lambda) = Regularization parameter
- $\beta i2$βi2 = Square of coefficients

Working

- Shrinks coefficient values but does not eliminate them.
- Reduces model variance.
- Handles multicollinearity effectively.

Advantages

- Prevents overfitting.
- Suitable when all features are important.

Lasso Regression (L1 Regularization)

Lasso Regression adds an L1 penalty to the loss function, which can shrink some coefficients exactly to zero.

Mathematical Formula

$Loss=SSE+\lambda\sum|\beta i|$Loss=SSE+λ∑|βi|

Working

- Performs automatic feature selection.
- Removes less important variables by making their coefficients zero.

Advantages

- Creates simpler and sparse models.
- Useful when dataset has many irrelevant features.

Implementation Steps on Real-World Dataset

1. Import Libraries
    - NumPy
    - Pandas

    - Scikit-learn
2. Load Dataset
    - Real-world dataset (e.g., housing prices, sales data, academic scores).
3. Data Preprocessing
    - Handle missing values.
    - Encode categorical variables.
    - Normalize/Standardize features.
4. Split Dataset
    - Training set
    - Testing set
5. Train Models
    - Multiple Linear Regression
    - Ridge Regression
    - Lasso Regression
6. Model Evaluation
    - R^2 Score
    - Mean Squared Error (MSE)
    - Cross-validation
7. Compare Performance
    - Analyze accuracy.
    - Observe coefficient values.
    - Select optimal lambda using cross-validation.

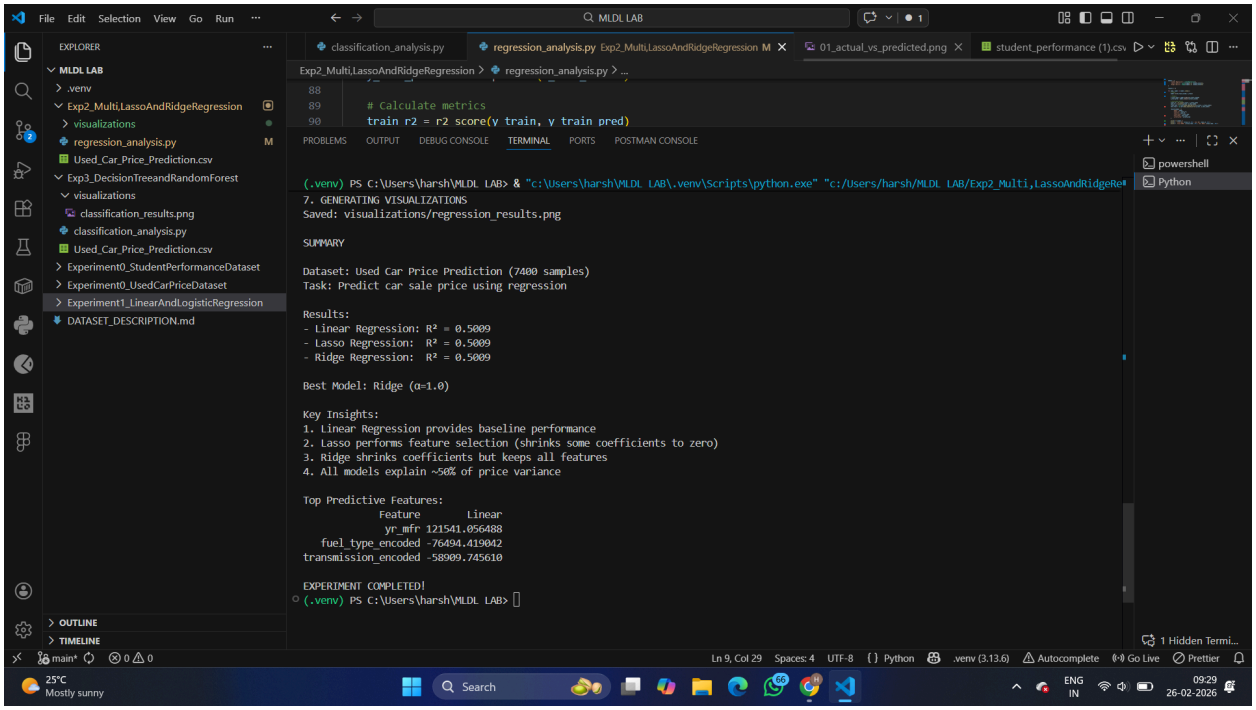
Code :

```
# Calculate metrics
train_r2 = r2_score(y_train, y_train_pred)
```

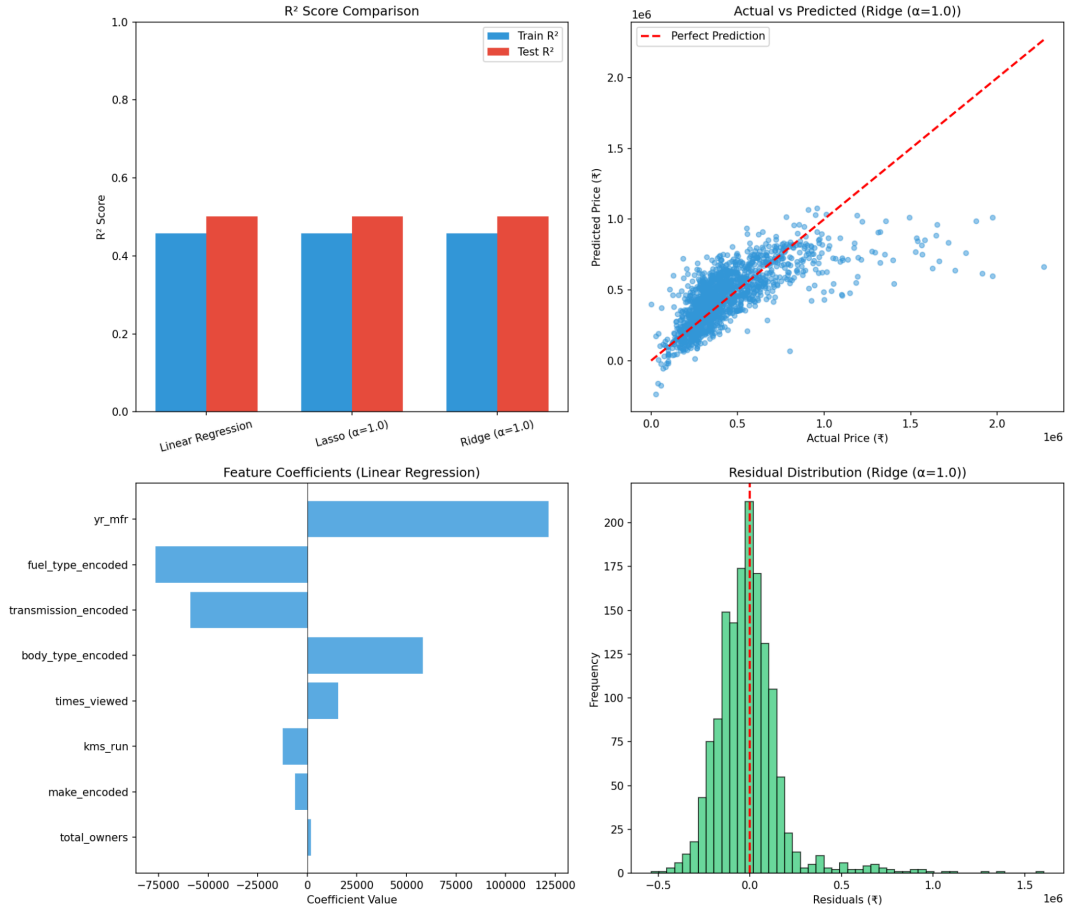
```
(.venv) PS C:\Users\harsh\MLDL LAB> & "c:\Users\harsh\MLDL LAB\.venv\Scripts\python.exe" "c:/Users/harsh/MLDL LAB/Exp2_Multi,LassoAndRidgeRe#
EXPERIMENT 2: MULTIPLE LINEAR, LASSO, AND RIDGE REGRESSION

1. LOADING DATA
Dataset Shape: (7400, 29)

2. PREPARING FEATURES
Features: ['yr_mfr', 'kms_run', 'times_viewed', 'total_owners', 'fuel_type_encoded', 'body_type_encoded', 'transmission_encoded', 'make_enco
ded']
X shape: (7400, 8), y shape: (7400,)

3. SPLITTING DATA
Training samples: 5920
Testing samples: 1480

4. TRAINING MODELS

Linear Regression:
  Train R²: 0.4583 | Test R²: 0.5009
  Train RMSE: ₹211,309 | Test RMSE: ₹186,402

Lasso (α=1.0):
  Train R²: 0.4583 | Test R²: 0.5009
  Train RMSE: ₹211,309 | Test RMSE: ₹186,402

Ridge (α=1.0):
  Train R²: 0.4583 | Test R²: 0.5009
  Train RMSE: ₹211,309 | Test RMSE: ₹186,400

5. MODEL COMPARISON
Model              Train R²     Test R²      Test RMSE
Linear Regression    0.4583      0.5009      ₹186,402
Lasso (α=1.0)        0.4583      0.5009      ₹186,402
```

```
# Calculate metrics
train_r2 = r2_score(y_train, y_train_pred)
```

```
(.venv) PS C:\Users\harsh\MLDL LAB> & "c:\Users\harsh\MLDL LAB\.venv\Scripts\python.exe" "c:/Users/harsh/MLDL LAB/Exp2_Multi,LassoAndRidgeRe#
5. MODEL COMPARISON
Model              Train R²     Test R²      Test RMSE
Linear Regression    0.4583      0.5009      ₹186,402
Lasso (α=1.0)        0.4583      0.5009      ₹186,402
Ridge (α=1.0)        0.4583      0.5009      ₹186,400

Best Model: Ridge (α=1.0) (Test R² = 0.5009)

6. FEATURE COEFFICIENTS
            Feature        Linear         Lasso         Ridge
             yr_mfr 121541.056488 121540.313110 121518.871833
           kms_run -12291.479865 -12290.339034 -12294.915793
       times_viewed  15531.140399  15530.331142  15529.456765
       total_owners   1838.147367   1836.789207   1831.844280
   fuel_type_encoded -76494.419042 -76493.281392 -76484.768758
   body_type_encoded  58259.121584  58258.389411  58256.980972
transmission_encoded -58909.745610 -58909.025232 -58902.731128
        make_encoded  -6068.417767  -6067.264731  -6063.819222

7. GENERATING VISUALIZATIONS
Saved: visualizations/regression_results.png

SUMMARY

Dataset: Used Car Price Prediction (7400 samples)
Task: Predict car sale price using regression

Results:
- Linear Regression: R² = 0.5009
- Lasso Regression:  R² = 0.5009
- Ridge Regression:  R² = 0.5009
```


File Edit Selection View Go Run
MLDL LAB
EXPLORER
MLDL LAB
.venv
Exp2_Multi,LassoAndRidgeRegression
visualizations
regression_analysis.py
Used_Car_Price_Prediction.csv
Exp3_DecisionTreeandRandomForest
visualizations
classification_results.png
classification_analysis.py
Used_Car_Price_Prediction.csv
Experiment0_StudentPerformanceDataset
Experiment0_UsedCarPriceDataset
Experiment1_LinearAndLogisticRegression
DATASET_DESCRIPTION.md
classification_analysis.py
regression_analysis.py Exp2_Multi,LassoAndRidgeRegression M
01_actual_vs_predicted.png
student_performance (1).csv
Exp2_Multi,LassoAndRidgeRegression > regression_analysis.py > ...
Calculate metrics
train r2 = r2_score(y_train, y_train_pred)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
(.venv) PS C:\Users\harsh\MLDL LAB> & "c:\Users\harsh\MLDL LAB\.venv\Scripts\python.exe" "c:/Users/harsh/MLDL LAB/Exp2_Multi,LassoAndRidgeRe
7. GENERATING VISUALIZATIONS
Saved: visualizations/regression_results.png
SUMMARY
Dataset: Used Car Price Prediction (7400 samples)
Task: Predict car sale price using regression
Results:
- Linear Regression: R² = 0.5009
- Lasso Regression: R² = 0.5009
- Ridge Regression: R² = 0.5009
Best Model: Ridge (α=1.0)
Key Insights:
1. Linear Regression provides baseline performance
2. Lasso performs feature selection (shrinks some coefficients to zero)
3. Ridge shrinks coefficients but keeps all features
4. All models explain ~50% of price variance
Top Predictive Features:
Feature Linear
yr_mfr 121541.056488
fuel_type_encoded -76494.419042
transmission_encoded -58909.745610
EXPERIMENT COMPLETED!
(.venv) PS C:\Users\harsh\MLDL LAB>
powershell
Python
OUTLINE
TIMELINE
1 Hidden Termi...
Ln 9, Col 29 Spaces: 4 UTF-8 Python .venv (3.13.6) Autocomplete Go Live Prettier
25°C
Mostly sunny
Search
ENG
IN
09:29
26-02-2026

Output :


R² Score Comparison
Train R²
Test R²
R² Score
Linear Regression
Lasso (α=1.0)
Ridge (α=1.0)
Actual vs Predicted (Ridge (α=1.0))
Perfect Prediction
Predicted Price (₹)
Actual Price (₹)
1e6
Feature Coefficients (Linear Regression)
yr_mfr
fuel_type_encoded
transmission_encoded
body_type_encoded
times_viewed
kms_run
make_encoded
total_owners
Coefficient Value
Residual Distribution (Ridge (α=1.0))
Frequency
Residuals (₹)
1e6

Conclusion :

Hence,Multiple Linear Regression is a good starting point because it helps us understand the basic relationship between features and the target variable, but it can sometimes overfit when the data is complex. Ridge Regression works better when features are highly correlated and we want to keep all of them in the model while reducing their impact slightly. Lasso Regression, on the other hand, is very useful when we want the model to automatically select the most important features and remove the less useful ones, making it simpler and more interpretable. Ultimately, the best choice depends on the nature of the dataset, the presence of multicollinearity, and the overall objective of the problem. Using cross-validation is essential to find the right regularization strength and ensure the model performs well on new data.